DHUM101 Essay (25 Marks)

Due Date: Thursday, 17 September, 9:00am. **Word Count**: 1500 words

For this essay, consider an emerging digital technology product type. Your job is to consider the technology and assess what ethical issues it raises. First, describe the technology briefly, spending no more than 1/5th of the essay (300 words) on that description. Then, discuss/analyse/report on its potential personal, social, cultural and environmental impacts (good and bad).

This assessment lists a set of questions and topics that you will want to consider when evaluating the technology. These are all areas that have been raised in lecture, tutorial or lab. While you want to consider all areas, you do not need to address them all in your essay. Some questions will be more relevant to certain technologies. For example, one technology may have a strong impact on the environment, while another technology more strongly affects privacy. Focus your essay on the important areas for your selected technology.

Imagine that you are writing your report for the rest of the DHUM101 class to critique, so you can assume they already know what has been discussed in class. Reviewing exercises from tutorial and lab will also help you prepare for this essay. We recommend choosing a technology of interest and then examining it with the provided topics.

Select a technology from this list.

- EFTPOS via biometrics
- Humanoid household robots
- Fully driverless vehicles
- Body Area Network
- AI traffic control
- AI companions for the elderly
- AI personalised learning at scale
- Brain-computer interface for members of the military/defense
- Schooling via an online 'metaverse'
- AI-controlled cloud-connected household appliances

Consider the following questions or topics

- Privacy. Consider issues of privacy. You may wish to particularly utilise the principles from New Zealand's privacy act.
- The autonomy of the individual as discussed in lecture. This may be broken down into two components (based on Carina Prunkl's work [covered in lab]).
    - Authenticity. The beliefs, values, motivations and reasons held by a person are in a relevant sense authentic to that person, i.e., not the product of external manipulative or distorting influences.

    - Agency. A person is able to act on the beliefs and values they hold. This implies that they have meaningful options available to them, allowing them to make choices that are of practical import to their life.
- The social impact, particularly on marginalised groups. You may consider many aspects but be sure to look for (a) biased data, (b) abusability, (c) dark rabbit holes.
- Sustainability. Consider the potential impacts that the technology could have on economic, environmental and social sustainability. It is not required to discuss any of these that you do not think will be realistically and significantly impacted.
- The life cycle of development. Consider the stages of planning, building, deployment, and monitoring. Where might you build in tasks to help improve the social impact of the tool?

Rubric

The lecturers will be using this rubric for assessment. You can also use it to guide your own essay development.

Marking Range	**Characteristics of the Essay**
80-100	Clearly and accurate presents (within the word constraints) a specific technology idea that can then be evaluated coherently. The essay has clearly identified key risks and stated them unambiguously. Those risks relate to the material taught in class and the questions/topics raised in this assessment. When presenting those risks, the essay demonstrates a robust understanding of both the topic and the risk, using all concepts accurately and richly. While there is evidence that all notions were considered, the essay focuses on those areas most relevant for the technology. There are no obvious gaps in discussion (such as no mention of privacy in a technology with immediate privacy impacts).
70-79	Indicates (within the word constraints) a specific technology idea though there are some ambiguities to what is being evaluated or its too general. The essay has identified key risks and stated them. Those risks relate to within the material taught in class and the questions/topics raised in this assessment. When presenting those risks, the essay demonstrates an understanding of both the topic and the risk, though the usage may be sparser with some ambiguities. The essay focuses on those areas most relevant for the technology, but there could be gaps so that it's not clear if important aspects were fully considered.
60-69	Indicates (within the word constraints) a specific technology idea but there are some inaccuracies or misunderstandings.

	The essay has identified risks and stated them. Those risks relate to the material taught in class and the questions/topics raised in this assessment. When presenting those risks, the essay demonstrates a mixed understanding of both the topic and the risk. The argument is not always clear to follow and may wander. Some areas might have been ignored without justification.
50-59	Indicates (within the word constraints) a technology idea but there are some inaccuracies or misunderstandings, or the idea is presented so generally that it has become very challenging to evaluate its risks. The essay has identified risks and stated them. Those risks are a mix of material taught in class and outside topics that are not the focus for the assessment. There is a surface understanding of the items being mentioned but little depth. Some areas might have been ignored without justification.
0-50	Indicates (within the word constraints) a technology idea but there are several inaccuracies or misunderstandings, or the idea is very general and amorphous making any evaluation impossible. The essay has identified risks and stated them. Those risks are a mix of material taught in class and outside topics that are not the focus for the assessment. Some parts might be written as if they are not using the class concepts at all, but “freewheeling it”. There is a surface understanding of the items being mentioned, or there may be direct mistakes in understanding. Some areas have been ignored without justification.